

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – TECHNICAL PRESENTATION

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO3 – Precision (BL3)	Assessment:	Assessment Tool 2 – Technical Presentation
Weightage:	30 % of CIA	Total Marks:	20 (Answer ANY ONE)
CO3 Statement:	<i>Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks.. (BTL 3)</i>		
Student Name:	P Reddy Swetha	Reg. No.:	192311206
Section / Batch:	Slot A	Date:	07-08-2026

Code of Conduct

I P Reddy Swetha – 192311206 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P Reddy Swetha

Instructions

- Duration: 10 minutes

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Questions

1. A CDMA network's near-far problem causes a near user's received power to be 20 dB higher than a far user's. Derive the resulting SIR degradation experienced by the far user, showing the dB arithmetic clearly. Explain in detail how power control resolves this problem by equalizing received power levels, and discuss why this derivation demonstrates the necessity of fast, closed-loop power control in CDMA.



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CO3 - AT2 Technical Presentation

P Reddy Swetha - 192311206

ITA0301 Mobile Computing

Course Faculty : Dr. Terrance Frederick Fernandez,
Professor, SIMATS Engineering, SIMATS

PROBLEM

- A CDMA network's near-far problem causes a near user's received power to be 20 dB higher than a far user's. Derive the resulting SIR degradation experienced by the far user, showing the dB arithmetic clearly. Explain in detail how power control resolves this problem by equalizing received power levels, and discuss why this derivation demonstrates the necessity of fast, closed-loop power control in CDMA.

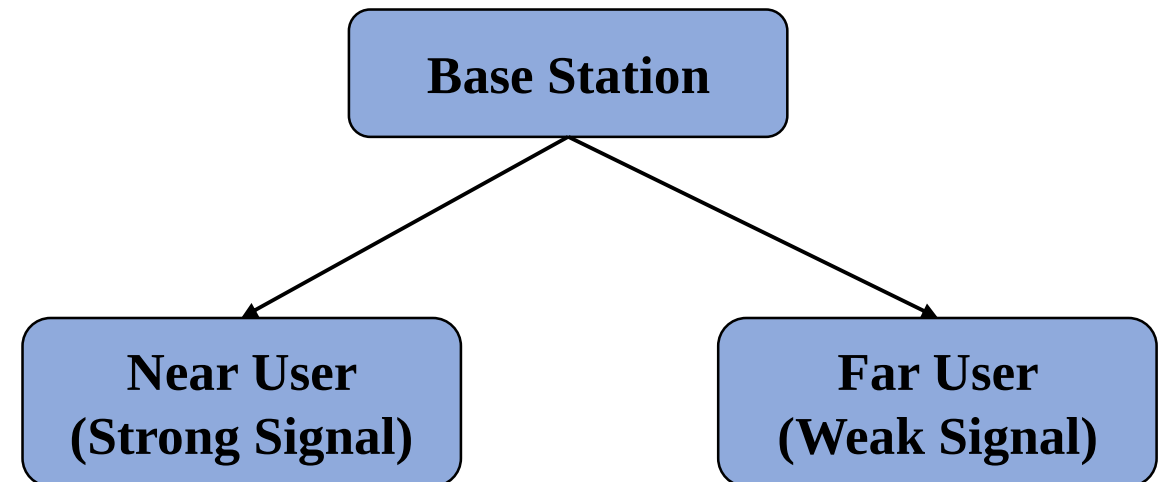
Near-Far Problem in a CDMA network

Near-Far Problem

- Occurs when users are at different distances from the base station.
- Near user's signal is stronger than the far user's signal.
- Strong signal interferes with the weak signal.

Given Condition

- Near User Received Power = **+20 dB**
- Far User Received Power = **0 dB**
- Power Difference = **20 dB**



SIR Degradation Calculation

- calculate how much the far user's Signal-to-Interference Ratio (SIR) decreases when the near user's received power is 20 dB higher.

Given

Near User Received Power = **+20 dB**

Far User Received Power = **0 dB**

Step 1: Convert dB to Power Ratio

$$\text{Power Ratio} = 10^{\frac{\text{dB}}{10}}$$

$$\text{Power Ratio} = 10^{\frac{20}{10}} = 100$$

- Near user's signal = 100 × stronger

Step 2: Calculate SIR

$$\text{SIR} = \frac{\text{Desired Signal}}{\text{Interference}}$$

$$= \frac{1}{100} = 0.01$$

$$\text{SIR} = 10 \log_{10}(0.01) = -20 \text{ dB}$$

- Far user's SIR = -20 dB

How power control resolves this problem

Power Control : Power Control is a technique used in CDMA to adjust the transmission power of every mobile user.

Without Power Control

- Near user : Strong signal
- Far user : Weak signal
- Strong signal interferes with weak signal
- Poor SIR

With Power Control

- Near user : Decrease transmit power
- Far user : Increase transmit power
- Equal received power at the Base Station
- Improved SIR

Need for Fast Closed-Loop Power Control

Power Control : Power Control is a technique used in CDMA to adjust the transmission power of every mobile user.

Why Fast?

- Users are constantly moving.
- Signal strength changes due to fading and distance.
- Quick power adjustment maintains signal quality.

Why Closed-Loop?

- Base station measures the received power.
- Sends Increase Power or Decrease Power commands.
- Mobile adjusts its transmit power continuously.

Conclusion

- The SIR calculation showed that a 20 dB power difference causes the far user's SIR to become -20 dB, which severely degrades communication.
- To prevent this situation, CDMA uses fast closed-loop power control, ensuring that all users are received with nearly equal power, reducing interference and maintaining reliable communication."



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THANK YOU



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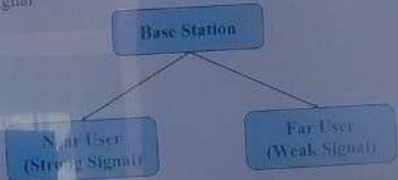


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      graph TD
        BS[Base Station] --- NU[Near User  
(Strong Signal)]
        BS --- FU[Far User  
(Weak Signal)]
      
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Received Power = -20 dB
 Transmitted Power = 0 dB
 Path Loss = 20 dB